

## Zack Andalman

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ADS; ArXiv

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### EDUCATION

|                      |               |                     |          |             |
|----------------------|---------------|---------------------|----------|-------------|
| Princeton University | Princeton, NJ | Ph.D., Astrophysics | 4.00 GPA | 2023 - 2028 |
| Yale University      | New Haven, CT | B.S., Physics       | 3.95 GPA | 2019 - 2023 |
| Evanston Township HS | Evanston, IL  | Diploma             | 4.00 GPA | 2015 - 2019 |

### PUBLICATIONS

#### First-Author

*Published: 3*

- [1] **Andalman, Z. L.**; Quataert, E.; Coughlin, E.; Nixon, C. (2026)  
*Resolving the (Debate About) Nozzle Shocks in Tidal Disruption Events*. OJA.  
<https://ui.adsabs.harvard.edu/abs/2025arXiv251208928A/abstract>
- [2] **Andalman, Z. L.**, Teyssier, R., & Avishai, D. (2025).  
*On the Origin of the High Star-Formation Efficiency in Massive Galaxies at Cosmic Dawn*. MNRAS.  
<https://ui.adsabs.harvard.edu/abs/2024arXiv241020530A/abstract>
- [3] **Andalman, Z. L.**; Liska, M. T. P.; Tchekhovskoy, A.; Coughlin, E. R.; & Stone, N. (2022).  
*Tidal Disruption Discs Formed and Fed by Stream-stream and Stream-disc Interactions in Global GRHD Simulations*. MNRAS.  
<https://ui.adsabs.harvard.edu/abs/2022MNRAS.510.1627A/abstract>

*In prep: 1*

- [4] **Andalman, Z. L.**; Fryer, C.; Fontes, C.; Mumpower, M.; Wolleager, R. (2026).  
*Thermalization in Kilonova Ejecta with Transport and Detailed Microphysics*.  
Planned submission to ApJ.

#### Co-Author

*Published: 1*

- [5] Kaaz, N.; et. al. incl. **Andalman, Z. L.** (2023).  
*Nozzle Shocks, Disk Tearing and Streamers Drive Rapid Accretion in 3D GRMHD Simulations of Warped Thin Disks*. MNRAS.  
<https://ui.adsabs.harvard.edu/abs/2023ApJ...955...72K/abstract>

*Submitted: 2*

- [6] Nixon, C.; Coughlin, E.; & **Andalman, Z. L.** (2026).  
*On the Origin of Anomalous Dissipation in Simulations of Tidal Disruption Events*. ApJ.  
<https://ui.adsabs.harvard.edu/abs/2026arXiv260520327N/abstract>
- [7] Sunseri, J.; **Andalman, Z. L.**; Teyssier, R. (2026).  
*Supermassive Black Hole Growth in Massive Galaxies at Cosmic Dawn*. MNRAS.  
<https://ui.adsabs.harvard.edu/abs/2025arXiv251019822S/abstract>

## **PRESENTATIONS**

### **Selected Talks**

|                                                                                |      |
|--------------------------------------------------------------------------------|------|
| TDEs and Nuclear Transients in Crete                                           | 2026 |
| Inaugural Tinsley workshop                                                     | 2024 |
| LANL Center for Nonlinear Studies Student Talk Series (1st place student talk) | 2024 |
| KITP Program - Towards a Physical Understanding of Tidal Disruption Events     | 2024 |
| RAMSES User Meeting                                                            | 2024 |
| HEAD Frontiers Seminar Series                                                  | 2023 |
| HEAD-19 Conference (invited)                                                   | 2022 |

### **Selected Posters**

|                                                        |            |
|--------------------------------------------------------|------------|
| AAS-245 Conference (Chambliss honorable mention)       | 2025       |
| HEAD-20 Conference (1st place undergraduate poster)    | 2023       |
| Connecticut Space Grant Consortium Expo                | 2021, 2022 |
| Blue Waters Symposium for Petascale Science and Beyond | 2018, 2019 |

## **SKILLS**

Computer languages: PYTHON, C, C++, FORTRAN, HTML/CSS/JAVASCRIPT, Unix shell

Software: H-AMR, RAMSES, ATHENA++/K, CLOUDY, MESA

Languages: English (native), Spanish (conversational), Modern Greek (basic)

Hobbies: Triathlon, jazz piano

## **GRANTS, FELLOWSHIPS, AND AWARDS**

|                                                                            |             |
|----------------------------------------------------------------------------|-------------|
| DOE Computational Science Graduate Fellowship, Krell Institute             | 2023 - 2027 |
| Martin Schwarzschild Fellowship, Princeton University (departmental award) | 2023 - 2025 |
| Michael Manzella Award, Yale University (leadership award)                 | 2023        |
| Lamat Fellowship, University of California Santa Cruz (REU)                | 2022        |
| Hahn Scholarship, Yale University (competitive undergrad research program) | 2019 - 2021 |
| Student Project Grant, NASA Connecticut Space Grant Consortium             | 2020        |
| First-Year Summer Research Fellowship, Yale University                     | 2020        |

## **LEADERSHIP EXPERIENCE**

|                                                                                           |             |
|-------------------------------------------------------------------------------------------|-------------|
| Thunch (thursday lunch) seminar series organizer                                          | 2024        |
| Yale Undergraduate Aerospace Association; Project Leader, Director of Projects, President | 2020 - 2023 |
| Yale Club Triathlon, Captain                                                              | 2021 - 2022 |

## **PROFESSIONAL SERVICE**

Number of papers refereed

MNRAS: 2

YouTube channel with scientific visualizations

<https://www.youtube.com>

## OUTREACH AND TEACHING

### Recurring

|                                                                            |             |
|----------------------------------------------------------------------------|-------------|
| Teaching fellow, Princeton University Preparatory Program                  | 2025 - 2027 |
| Research mentor for the Princeton Undergraduate Summer Research Program    | 2026        |
| Teaching assistant, AST 206: Black Holes                                   | 2024 - 2026 |
| Tutor, New Jersey Prison Teaching Initiative                               | 2024        |
| Organizer, multiple aerospace-themed educational events with local schools | 2021        |

### One-time

|                                                                                 |             |
|---------------------------------------------------------------------------------|-------------|
| Volunteer, Princeton Spring into Science (educational event with local schools) | 4/16, 2026  |
| Speaker, Planetarium at New Jersey State Museum                                 | 2/21, 2026  |
| Speaker, Astronomy on Tap (Trenton, NJ)                                         | 11/20, 2024 |
| Speaker, solar eclipse at Littlebrook Elementary                                | 4/08, 2024  |
| Teacher, Yale Splash (annual event with local schools)                          | 4/16, 2022  |